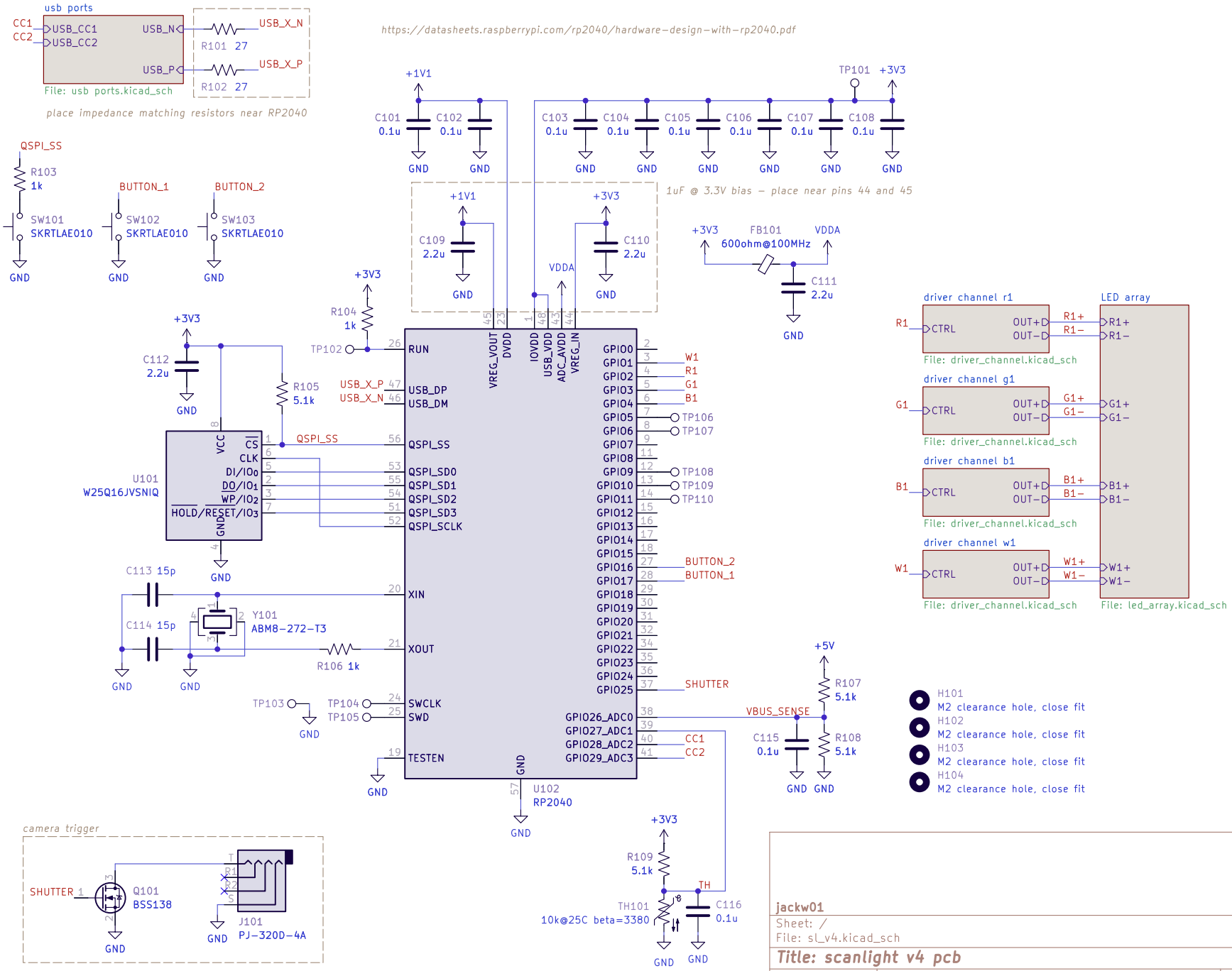
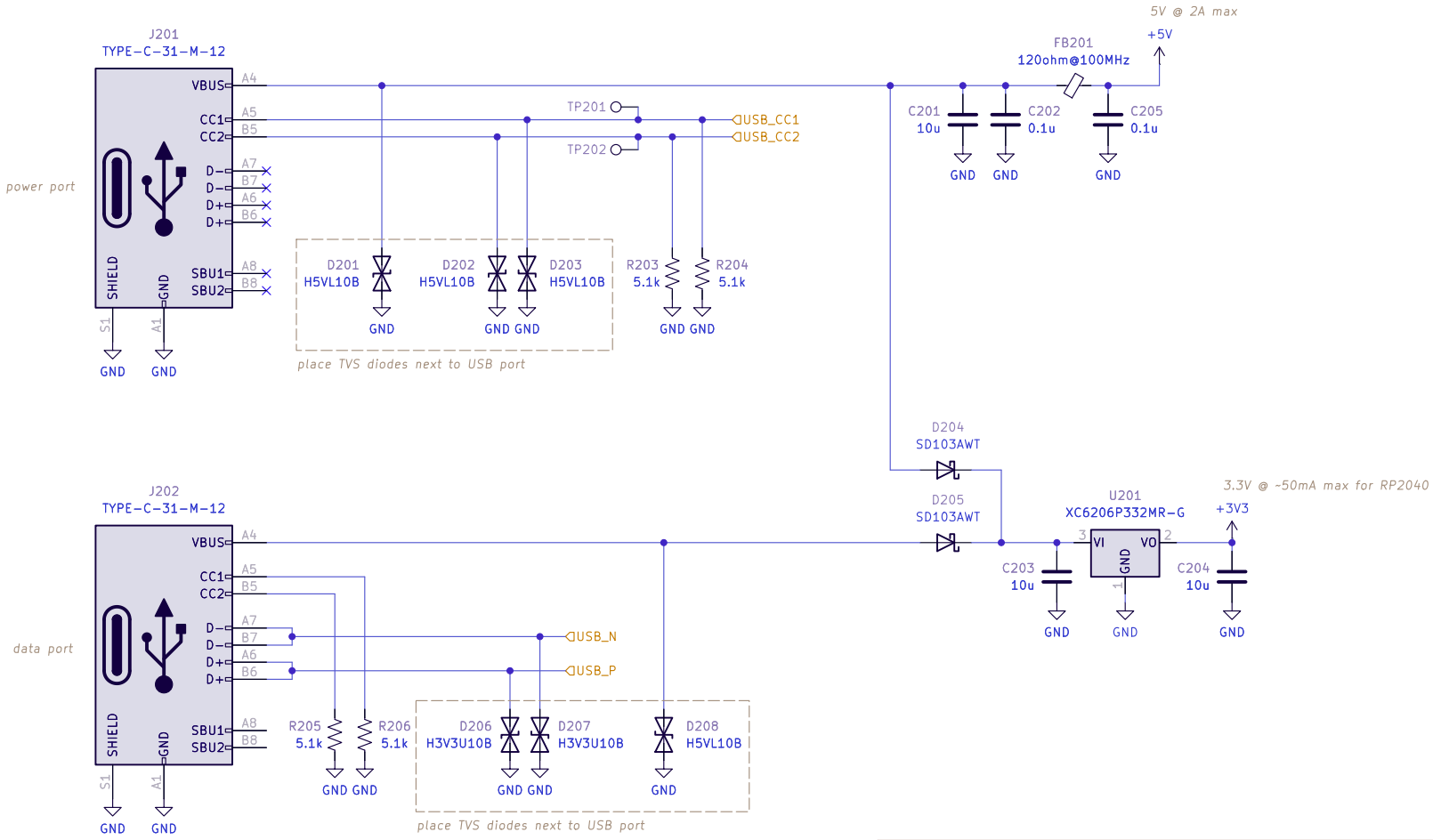
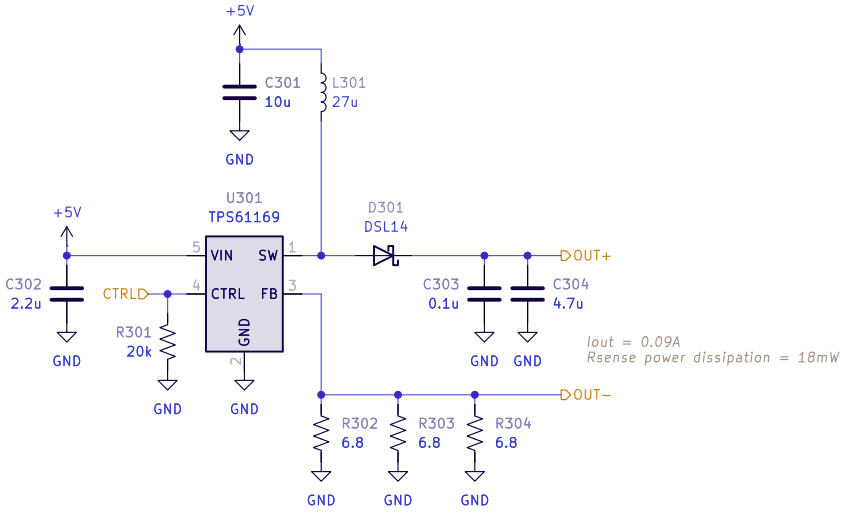
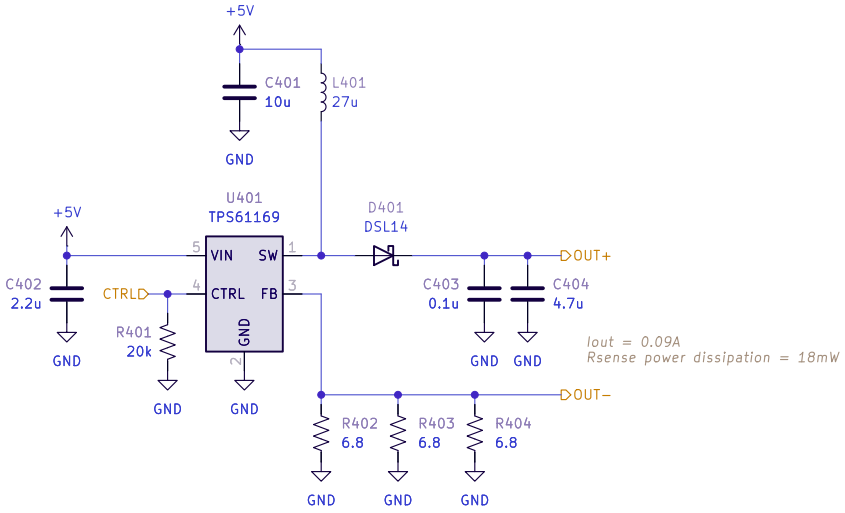

usb ports
CC1
CC2
USB_CC1
USB_CC2
USB_N
USB_P
USB_X_N
USB_X_P
R101 27
R102 27
File: usb ports.kicad_sch
place impedance matching resistors near RP2040
https://datasheets.raspberrypi.com/rp2040/hardware-design-with-rp2040.pdf
+1V1
+3V3
TP101
C101 0.1u
C102 0.1u
C103 0.1u
C104 0.1u
C105 0.1u
C106 0.1u
C107 0.1u
C108 0.1u
GND
1uF @ 3.3V bias — place near pins 44 and 45
C109 2.2u
C110 2.2u
VDDA
FB101
600ohm@100MHz
C111 2.2u
QSPI_SS
R103 1k
BUTTON_1
BUTTON_2
SW101 SKRTLAE010
SW102 SKRTLAE010
SW103 SKRTLAE010
R104 1k
TP102
C112 2.2u
R105 5.1k
U101
W25Q16JVSNIQ
CS
CLK
DI/IO0
DO/IO1
WP/IO2
HOLD/RESET/IO3
VCC
GND
C113 15p
C114 15p
Y101
ABM8-272-T3
R106 1k
U102
RP2040
RUN
USB_DP
USB_DM
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
XIN
XOUT
SWCLK
SWD
TESTEN
VREG_VOUT
DVDD
IOVDD
USB_VDD
ADC_AVDD
VREG_IN
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
GPIO29_ADC3
W1
R1
G1
B1
TP106
TP107
TP108
TP109
TP110
BUTTON_2
BUTTON_1
SHUTTER
CC1
CC2
TP103
TP104
TP105
driver channel r1
driver channel g1
driver channel b1
driver channel w1
CTRL
OUT+
OUT-
R1+
R1-
G1+
G1-
B1+
B1-
W1+
W1-
File: driver_channel.kicad_sch
LED array
File: led_array.kicad_sch
+5V
R107 5.1k
R108 5.1k
C115 0.1u
VBUS_SENSE
H101
H102
H103
H104
M2 clearance hole, close fit
camera trigger
SHUTTER
Q101
BSS138
J101
PJ-320D-4A
R109 5.1k
TH
TH101
10k@25C beta=3380
C116 0.1u
jackw01
Sheet: /
File: sl_v4.kicad_sch
Title: scanlight v4 pcb
Size: USLetter
Date: 2026-03-29
Rev: 1.0
KiCad E.D.A. 10.0.7
Id: 1/16



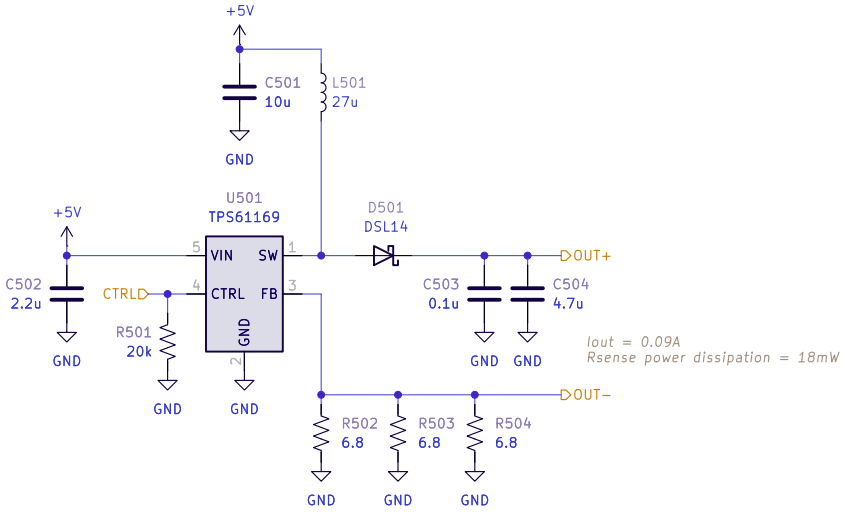
jackw01	
Sheet: /usb ports/	
File: usb ports.kicad_sch	
Title: scanlight v4 pcb	
Size: USLetter | Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7	Id: 2/16



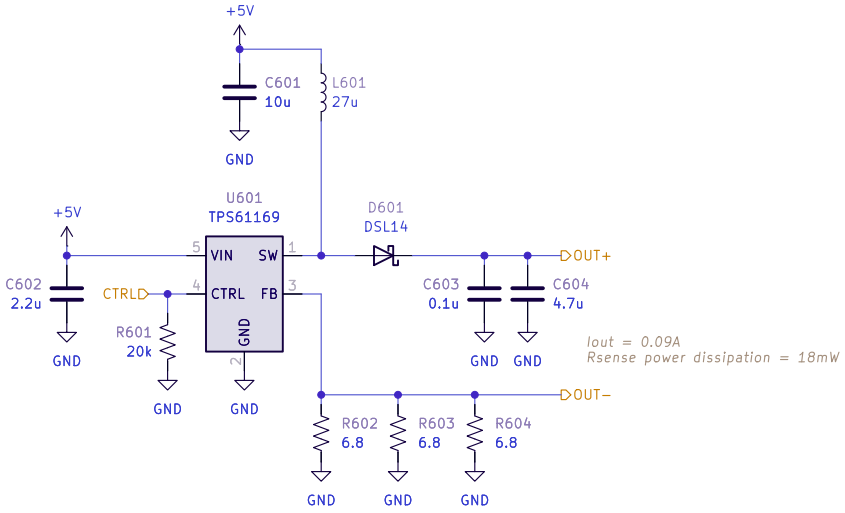
jackw01		
Sheet: /driver channel r1/		
File: driver_channel.kicad_sch		
Title: scanlight v4 pcb		
Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 3/16



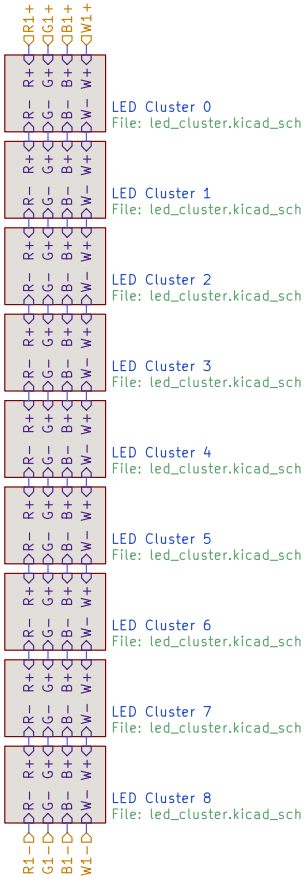
jackw01		
Sheet: /driver channel g1/		
File: driver_channel.kicad_sch		
Title: scanlight v4 pcb		
Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 4/16



jackw01		
Sheet: /driver channel b1/		
File: driver_channel.kicad_sch		
Title: scanlight v4 pcb		
Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 5/16



jackw01		
Sheet: /driver channel w1/		
File: driver_channel.kicad_sch		
Title: scanlight v4 pcb		
Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 6/16



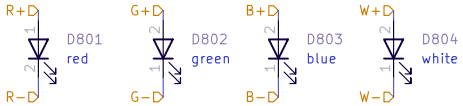
jackw01

Sheet: /LED array/

File: led_array.kicad_sch

Title: scanlight v4 pcb

Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 7/16



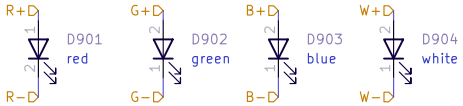
jackw01

Sheet: /LED array/LED Cluster 0/

File: led_cluster.kicad_sch

Title: scanlight v4 pcb

Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 8/16



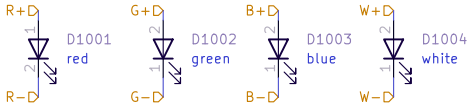
jackw01

Sheet: /LED array/LED Cluster 1/

File: led_cluster.kicad_sch

Title: scanlight v4 pcb

Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 9/16



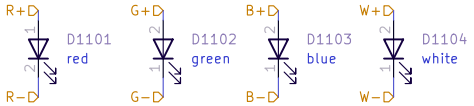
jackw01

Sheet: /LED array/LED Cluster 2/

File: led_cluster.kicad_sch

Title: scanlight v4 pcb

Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 10/16



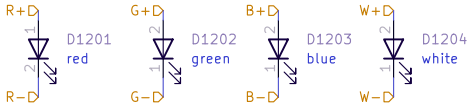
jackw01

Sheet: /LED array/LED Cluster 3/

File: led_cluster.kicad_sch

Title: scanlight v4 pcb

Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 11/16



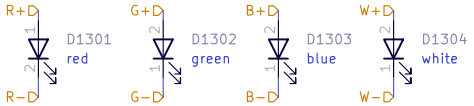
jackw01

Sheet: /LED array/LED Cluster 4/

File: led_cluster.kicad_sch

Title: scanlight v4 pcb

Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 12/16



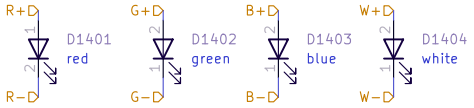
jackw01

Sheet: /LED array/LED Cluster 5/

File: led_cluster.kicad_sch

Title: scanlight v4 pcb

Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 13/16



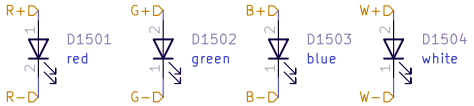
jackw01

Sheet: /LED array/LED Cluster 6/

File: led_cluster.kicad_sch

Title: scanlight v4 pcb

Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 14/16



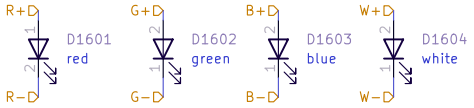
jackw01

Sheet: /LED array/LED Cluster 7/

File: led_cluster.kicad_sch

Title: scanlight v4 pcb

Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 15/16



jackw01

Sheet: /LED array/LED Cluster 8/

File: led_cluster.kicad_sch

Title: scanlight v4 pcb

Size: USLetter	Date: 2026-03-29	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 16/16